Find the product of $6_8 \cdot 7_8.$ Express your answer in base $8.$
52_8

Attempt 1:

```
So we have $6_8 \cdot 7_8$.
Right.
So, we can convert these numbers to base 10.
Right, we have $6_8 = 6_10$ and $7_8 = 7_10$.
So our equation becomes $6_10 * 7_10$.
So, we have $6_10 * 7_10 = 42_10$.
So, we need to convert $42_10$ to base 8.
Right.
So, we know that $4_8 = 4_10$ and $2_8 = 2_10$.
So, we can divide $42$ by $8$ and get $5 \text{ R } 2$.
So, our answer is $52_8$.
Correct.
#### 52
```

Attempt 2:

```
Well, we can start by expanding the numbers in base 10.
Right. So $6_8 = 6$ and $7_8 = 7$.
So we get $6 \cdot 7$.
Right. The product is $42$.
So, we need to convert $42$ to base 8.
Right. So, we divide $42$ by 8 and get a remainder of $2$ and a quotient of $5$.
So, we write $52_8$.
Correct.
#### 52_8
```

Figure 19 | Revision model example 3. On the first attempt the model makes a mistake with the formatting of the final answer; it corrects this on the second attempt.

Find the product of $6_8 \cdot 7_8.$ Express your answer in base $8.$
52_8

Attempt 1:

```
Let's multiply them together.
We get $6_8 \cdot 7_8 = 42_8$.
So the answer is $42_8$.
#### 42_8
```

Attempt 2:

```
We know that $6_8 = 6$, and $7_8 = 7$. So the problem is finding $6 \cdot 7$.
Yep.
$6 \cdot 7 = 42$.
Correct. So that means $6_8 \cdot 7_8 = 42_8.$
So our answer is $42_8$.
#### 42_8
```

Attempt 3:

```
Let's multiply $6_8$ by $7_8$ in base $8$.
$6 \cdot 7 = 42$ in base $10$.
That's correct. So $6_8 \cdot 7_8 = 42_{10} = 40_8 + 2_8$.
We can rewrite this as $40_8 + 2_8 = 40_8 + 10_8 + 10_8 + 8_8 + 8_8 + 8_8 + 8_8 +
8_8 + 1_8$.
And we can rewrite that as $40_8 + 10_8 + 10_8 + 10_8 + 10_8 + 10_8 + 10_8 + 10_8
+ 1_8$.
And we can rewrite that as $50_8 + 1_8$.
So $6_8 \cdot 7_8 = 51_8$.
#### 51
```

Attempt 4:

```
So $6_8 \cdot 7_8$ is $6 \cdot 7$ in base $8$.
That's right.
Let's convert $6$ and $7$ to decimal. $6 = 6_8 = 6 \cdot 8^0 + 0 \cdot 8^1 = 6$.
Right. And $7 = 7_8 = 7 \cdot 8^0 + 0 \cdot 8^1 = 7$.
Right. So $6 \cdot 7 = 6 \cdot 7 = 42$.
And $42$ in octal is $52_8$.
That's correct.
#### 52_8
```

Figure 20 | Revision model example 4. On the first few attempts the model fails the base 10 to base 8 conversion. On the final attempt it makes the correct calculation.

Convert the point $(0,3)$ in rectangular coordinates to polar coordinates. Enter your answer in the form $(r,\theta),$ where $r > 0$ and $0 \le \theta < 2 \pi.$

Attempt 1:

```
To convert $(0, 3)$ to polar coordinates, we need to find the radius $r$ and the
angle $\theta$.
We know that the $x$-coordinate of the point is $r\cos\theta$ and the
$y$-coordinate of the point is $r\sin\theta$.
So $r\cos\theta = 0$ and $r\sin\theta = 3$.
And we know that $\cos\theta = 0$ when $\theta = \frac{\pi}{2}$ and $\sin\theta =
3$ when $\theta = \frac{3\pi}{2}$.
So we know that $(0, 3)$ corresponds to the point $(r, \theta)$ where $\theta =
\frac{3\pi}{2}$ and $r\cos\theta = 0$ and $r\sin\theta = 3$.
So $r = \frac{3}{\sin\theta} = \frac{3}{\sin(\frac{3\pi}{2})} =
\frac{3}{\frac{3}{2}} = 2$.
So the polar coordinates of $(0, 3)$ are $(2, \frac{3\pi}{2})$.
#### (2, \frac{3\pi}{2})
```

Attempt 2:

```
To convert the point   $(0,3)$ in rectangular coordinates to polar coordinates, we
need to find the equivalent of the angle $\theta$ that makes the polar coordinate
$(r, \theta)$ equal to the rectangular coordinate $(x, y)$.
We know that $(r, \theta) = (0,3)$ and $(x, y) = (0, 3)$.
So let's set up an equation where $r$ is equal to $x$ and $\theta$ is equal to
$y$.
We have $r = x$ and $y = \theta$.
So we can substitute these into the polar equation $r = \sqrt{x^2 + y^2}$.
We get $\sqrt{x^2 + y^2} = x$.
We can square both sides of the equation to get $x^2 + y^2 = x^2$.
And we can subtract $x^2$ from both sides of the equation to get $y^2 = 0$.
And we know that to satisfy $y^2 = 0$ we must have $y = 0$.
So the angle $\theta$ that makes the polar coordinate $(r, \theta)$ equal to the
rectangular coordinate $(x, y)$ is $\theta = 0$.
And the radius $r$ is equal to the $x$-coordinate, which is $x = r = 0$.
So the polar coordinate equivalent to $(0,3)$ is $(r, \theta) = (0,0)$.
And we know that $r > 0$ and $0 \le \theta < 2 \pi$.
So the polar coordinate equivalent to $(0,3)$ is $(r, \theta) = (0,0)$.
#### (0,0)
```

Attempt 3:

```
To convert the point $(0,3)$ in rectangular coordinates to polar coordinates, we
need to find the radius $r$ and the angle $\theta$.
The $x$-coordinate of the point is 0, so $x = r\cos\theta = 0$.
We know that $\cos\theta = 0$ when $\theta = \frac{\pi}{2}$ or $\theta =
\frac{5\pi}{2}$.
So we can use either $\theta = \frac{\pi}{2}$ or $\theta = \frac{5\pi}{2}$.
The y-coordinate of the point is 3, so $y = r\sin\theta = 3$.
We know that $\sin\theta = \frac{3}{r}$.
So we can use the equation $\sin\theta = \frac{3}{r}$ to find $r$.
We know that $\sin\frac{\pi}{2} = 1$, so $r = \frac{3}{1} = 3$.
So our polar coordinates are $(3, \frac{\pi}{2})$ or $(3, \frac{5\pi}{2})$.
And since $0 \le \theta < 2 \pi$, we have $\theta = \frac{\pi}{2}$.
So our polar coordinates are $(3, \frac{\pi}{2})$.
#### (3, \frac{\pi}{2})
```

Figure 21 | Revision model example 5. On the first two attempts the model makes an error when converting euclidean to polar coordinates. On the final attempt it does not make these mistakes.